

A RPS: Measurement and Definitions

A.1 Sample Restrictions

The Qualtrics panel includes about 15 million members and is not a random sample of the U.S. population. However, researchers can instruct Qualtrics to target survey invitations to specific demographic groups. The RPS sample was designed to be nationally representative of the U.S. across gender, age, education, household income, and other demographic characteristics. Individuals who take the survey too fast, i.e. in less than 50% of the median time in a soft launch, or who do not state that they will provide their best answers, are automatically dropped from the sample.¹ Finally, once fielding is completed Qualtrics staff goes over all responses and filters out any that look suspicious. This is typically the case for around 10% of the responses. We also screened manually out responses whose responses to the free text question about the industry appear to be generated by a bot taking the survey. This was the case for less than 10 responses in each survey.

The pilot survey in June 2024 received 2,551 responses. Our full surveys in August and November 2024 received 5,014 and 5,329 responses, respectively. In each survey, we dropped less than 40 respondents because they either reported their industry and/or occupation as military or because they reported being employed but also reported being homemakers, retired, or unemployed as their occupation. Table A.1 show the raw sample composition for the pooled August and November 2024 surveys after these observations were dropped.

We also include occupation in our weighting scheme. This requires us to drop another 108, 112, and 113 observations due to missing occupation codes for the June, August, and November surveys, respectively. We drop another 70, 154, and 145 employed respondents after constructing weights because we lack information on their Generative AI use at work last week. All of these dropped respondents were classified as “employed, absent from work last week” as they by construction cannot have used Generative AI last week. Accounting for all individuals dropped from our analysis, this leaves us with a final sample size of 94.7%, 96.8%, and 96.9% of the initially collected responses for the June and August surveys, respectively.

Table A.1 compares the sample composition between the CPS and RPS along the demographics targeted in the sampling procedure for our main surveys (columns 1 and 2). The most notable discrepancies are that individuals aged 18 to 24 and with no more than a high school

¹The exact phrasing of the screener question is: “We care about the quality of our survey data and hope to receive the most accurate measure of your opinions, so it is important to us that you thoughtfully provide your best answer to each question in the survey. Do you commit to providing your thoughtful and honest answers to the questions in this survey?” with the following three answer options (1) I will provide my best answers, (2) I will not provide my best answers, and (3) I can’t promise either way. According to Qualtrics staff, this question provides the best results in terms of screening out respondents.

degree are underrepresented in the RPS relative to the CPS, while individuals with household income of \$50,000 or less are overrepresented. The bottom panel of Table A.1 compares employment status in the CPS and RPS, statistics that have not been targeted in the sampling procedure. Employment rates are very similar across the two surveys, although individuals classified as unemployed according to the CPS definition are somewhat overrepresented in the RPS.

A.2 Weighting

As described in the body of the paper, we asked Qualtrics to administer the survey to a sample of respondents who match the U.S. population along a few broad demographic characteristics: gender, five age bins (18-24, 25-34, 35-44, 45-54, 55-64), race and ethnicity (non-Hispanic White, non-Hispanic Black, Hispanic, other), education (high school or less, some college or associate degree, bachelor’s degree or more), marital status (married or not), number of children in the household (0, 1, 2, 3 or more), three income bins for household income over the last 12 months (<\$50k, \$50k-100k, >\$100k), and four Census regions. Table A.1 compare the sample composition between the pooled August and November RPS and the CPS along the demographics targeted in the sampling procedure for our main survey. The tables look very similarly for each survey wave separately, including the June pilot, and are available upon request. As discussed in the paper, we also compare at the bottom of the table employment status in the CPS and RPS, as well as the demographic composition for the employed. None of this latter sets of moments were targeted during the sampling of survey respondents.

Using the iterative proportional fitting (raking) algorithm of Deming and Stephan (1940), we construct sampling weights to ensure the RPS matches the CPS sample proportions for the same set of demographic characteristics included in the Qualtrics sampling targets for the overall sample, i.e., independent of employment status. However, we use more disaggregated categories for education and marital status, and we interact all categories with gender. In particular, for education, we distinguish between less than high school, high school graduate or equivalent, some college but no degree, associate degree, bachelor’s degree, and graduate degree. For marital status, we distinguish between married + spouse present, divorced, never married, and "other." We also condition on relationship status (spouse living in the same household, partner living in the same household, other).

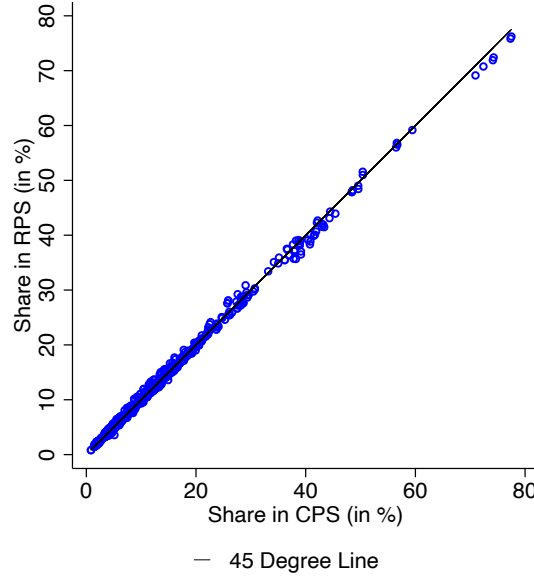
In addition, our sampling weights replicate the employed-at-work rates, the employment rates, and the labor force participation rates in each of the subsequent months. We match these key labor market statistics not only in the aggregate but also conditional on demographic characteristics. More specifically, we match the employed-at-work rate, the employment rate, and the labor force participation rate for the current month by gender, age (18-24, 25-34, 35-

Table A.1: Sample Composition in the Pooled August and November 2024 CPS and RPS

	<i>Everyone</i>		<i>Employed</i>	
	CPS	RPS	CPS	RPS
	(1)	(2)	(3)	(4)
<i>Gender: Women</i>	50.4	52.2	47.3	47.5
<i>Age</i>				
18-24	15.0	11.7	12.0	12.0
25-34	22.2	22.0	23.9	23.8
35-44	22.2	23.8	24.4	25.0
45-54	20.1	21.2	21.7	21.7
55-64	20.6	21.2	17.9	17.5
<i>Race/Ethnicity</i>				
Non-hispanic White	56.5	56.4	57.8	57.8
Non-hispanic Black	12.9	13.2	12.1	12.3
Hispanic	20.5	20.4	20.2	19.8
Other	10.1	10.0	9.9	10.1
<i>Education</i>				
Highschool or less	37.1	32.9	32.8	27.1
Some college/Associate's degree	25.8	27.6	25.2	27.6
Bachelor's or Graduate degree	37.1	39.5	42.0	45.3
<i>Marital Status: Married</i>	50.1	49.0	52.8	52.4
<i>Number of children</i>				
0	58.1	55.7	57.2	52.8
1	17.8	19.7	18.1	21.2
2	14.7	16.5	15.5	18.4
3+	9.3	8.1	9.2	7.6
<i>Household Income in Last 12 Months</i>				
\$0-\$50,000	26.1	31.4	19.9	22.6
\$50,000-\$100,000	29.9	29.5	30.7	32.1
\$100,000+	44.0	39.1	49.5	45.3
<i>Region</i>				
Northeast	17.0	18.3	16.9	18.8
Midwest	20.3	19.4	21.1	19.2
South	38.8	38.1	38.1	37.9
West	23.8	24.2	23.9	24.2
<i>Employment Status</i>				
Employed, at work last week	71.7	69.8		
Employed, absent from work last week	2.5	2.9		
Unemployed	3.2	8.0		
Not in the labor force	22.6	19.3		
<i>Observations</i>	115477	10241	85546	7450

Notes: Column 1 reports the sample composition in the pooled August and November 2024 Current Population Survey (CPS) for the variables targeted by Qualtrics in the sampling procedure. The employment status was the only variable not targeted. Column 2 reports the sample composition in the pooled August and November 2024 Real-Time Population Survey (RPS). The sample in both data sets is restricted to the civilian population ages 18-64. Columns 3 and 4 report the same outcomes for the employed (at work and absent from work last week).

FIGURE A.1: Sample Composition in the Weighted RPS vs. CPS for August and November 2024



Notes: The figure shows for all statistics used in the weighting scheme, the weighted fraction of individuals with the respective characteristics in the RPS (on the y-axis) and the CPS (on the x-axis).

44, 45-54, 55-64), race and ethnicity (non-Hispanic White, non-Hispanic Black, Hispanic, all other racial and ethnic groups), education (high school or less, some college or associate degree, bachelor's degree or more), marital status (married + spouse present, never married, other), relationship status (spouse living in the same household, partner living in the same household, other), presence of children in the household (yes or no), household income over the last 12 months (<\$50k, \$50k-100k, >\$100k), and region (Midwest, Northeast, South, and West using the Census definition). These groupings were chosen to ensure that each cell size is at least 30. We also include 2-digit occupation codes in our weighting scheme. Among the 22 occupations, we merge several occupations to ensure a sufficient sample size. In particular, for the June 2024 survey, we merge a) "Architecture and Engineering Occupations" and "Life, Physical, and Social Science Occupations", b) "Community and Social Service Occupations" and "Legal Occupations", c) "Healthcare Support Occupations" and "Protective Service Occupations", and d) "Farming, Fishing, and Forestry Occupations" and "Construction and Extraction Occupations." For the August and November 2024 survey, we proceeded similarly but, due to the larger sample size, did not need to merge "Architecture and Engineering Occupations" and "Life, Physical, and Social Science Occupations."

Including all the interaction terms, we have a total of 48 statistics (e.g., gender is one, and gender x education is another) which we weight on, with a combined 295 categories (e.g., gender has two categories, and gender x education has $2 \times 6 = 12$ categories; for the June 2024 we only have 294 categories because of the additional occupational grouping). To visualize the goodness